MPG No. 19.1.68 $3m$ (3m1 setting) [Type I, trigonal] [T tensor]

* Rank 0 tensor.

$$[T]$$

$$T = \mathbb{T}_0^{(1)}$$

* Rank 1 tensor.

$$\begin{bmatrix} 0 & 0 & T_z \end{bmatrix}$$

$$T_z = \mathbb{T}_z^{(1)}$$

* Rank 2 tensor (s).

$$\begin{bmatrix} T_{xx} & 0 & 0 \\ 0 & T_{xx} & 0 \\ 0 & 0 & T_{zz} \end{bmatrix}$$

$$T_{xx} = \mathbb{T}_0^{(1)} - \mathbb{T}_u^{(1)}$$
$$T_{zz} = \mathbb{T}_0^{(1)} + 2\mathbb{T}_u^{(1)}$$

* Rank 2 tensor (a). * Rank 3 tensor (ss).

$$\begin{bmatrix} 0 & T_{xxy} & T_{xxz} \\ 0 & -T_{xxy} & T_{xxz} \\ 0 & 0 & T_{zzz} \\ 0 & T_{xxz} & 0 \\ T_{xxz} & 0 & 0 \\ T_{xxy} & 0 & 0 \end{bmatrix}$$

$$T_{xxy} = \mathbb{T}_{3s}^{(1)}$$
$$T_{xxz} = \mathbb{T}_z^{(1)} - \mathbb{T}_z^{\alpha(1)}$$
$$T_{zzz} = 3\mathbb{T}_z^{(1)} + 2\mathbb{T}_z^{\alpha(1)}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & 0 & T_{xxz} \\ 0 & 0 & T_{xxz} \\ 0 & 0 & 0 \\ 0 & -\frac{T_{xxz}}{2} & 0 \\ -\frac{T_{xxz}}{2} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$T_{xxz} = 2\mathbb{T}_z^{(2)}$$
$$\frac{T_{xxz}}{2} = \mathbb{T}_z^{(2)}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} 0 & T_{yzy} & 0 \\ -T_{yzy} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$T_{yzy} = \mathbb{T}_z^{(3)}$$

* Rank 4 tensor (sss).

$$\begin{bmatrix} T_{xxxx} & T_{xxyy} & T_{xxzz} & T_{xxyz} & 0 & 0 \\ T_{xxyy} & T_{xxxx} & T_{xxzz} & -T_{xxyz} & 0 & 0 \\ T_{xxzz} & T_{xxzz} & T_{zzzz} & 0 & 0 & 0 \\ T_{xxyz} & -T_{xxyz} & 0 & T_{yzyz} & 0 & 0 \\ 0 & 0 & 0 & 0 & T_{yzyz} & T_{xxyz} \\ 0 & 0 & 0 & 0 & T_{xxyz} & \frac{T_{xxxx}}{2} - \frac{T_{xxyy}}{2} \end{bmatrix}$$

$$T_{xxxx} = \mathbb{T}_{0}^{(1)} + 2\mathbb{T}_{0}^{(2)} + 3\mathbb{T}_{40}^{(1)} - 2\mathbb{T}_{u}^{(1)} - 4\mathbb{T}_{u}^{(2)}$$
$$T_{xxyy} = \mathbb{T}_{0}^{(1)} + \mathbb{T}_{40}^{(1)} - 2\mathbb{T}_{u}^{(1)}$$
$$T_{xxzz} = \mathbb{T}_{0}^{(1)} - 4\mathbb{T}_{40}^{(1)} + \mathbb{T}_{u}^{(1)}$$
$$T_{xxyz} = \mathbb{T}_{4s}^{(1)}$$
$$T_{zzzz} = \mathbb{T}_{0}^{(1)} + 2\mathbb{T}_{0}^{(2)} + 8\mathbb{T}_{40}^{(1)} + 4\mathbb{T}_{u}^{(1)} + 8\mathbb{T}_{u}^{(2)}$$
$$T_{yzyz} = \mathbb{T}_{0}^{(2)} - 4\mathbb{T}_{40}^{(1)} + \mathbb{T}_{u}^{(2)}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & 0 & T_{xxzz} & T_{xxyz} & 0 & 0 \\ 0 & 0 & T_{xxzz} & -T_{xxyz} & 0 & 0 \\ -T_{xxzz} & -T_{xxzz} & 0 & 0 & 0 & 0 \\ -T_{xxyz} & T_{xxyz} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -T_{xxyz} \\ 0 & 0 & 0 & 0 & T_{xxyz} & 0 \end{bmatrix}$$

$$T_{xxzz} = 3\mathbb{T}_{u}^{(3)}$$
$$T_{xxyz} = 2\mathbb{M}_{3c}^{(1)}$$

* Rank 4 tensor (aas).

$$\begin{bmatrix} T_{yzyz} & 0 & 0 \\ 0 & T_{yzyz} & 0 \\ 0 & 0 & T_{xyxy} \end{bmatrix}$$

$$T_{yzyz} = \mathbb{T}_{0}^{(3)} - 2\mathbb{T}_{u}^{(4)}$$
$$T_{xyxy} = \mathbb{T}_{0}^{(3)} + 4\mathbb{T}_{u}^{(4)}$$

* Rank 4 tensor (aaa). * Rank 4 tensor (sa).

$$\begin{bmatrix} T_{xxyz} & 0 & 0 \\ -T_{xxyz} & 0 & 0 \\ 0 & 0 & 0 \\ T_{yzyz} & 0 & 0 \\ 0 & -T_{yzyz} & 0 \\ 0 & -T_{xxyz} & 0 \end{bmatrix}$$

$$T_{xxyz} = \mathbb{M}_{3c}^{(2)}$$
$$T_{yzyz} = -3\mathbb{T}_{u}^{(5)}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} T_{yzxx} & -T_{yzxx} & 0 & T_{yzyz} & 0 & 0 \\ 0 & 0 & 0 & 0 & -T_{yzyz} & -T_{yzxx} \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$T_{yzxx} = \mathbb{M}_{3c}^{(3)}$$
$$T_{yzyz} = -3\mathbb{T}_{u}^{(6)}$$

* Rank 4 tensor (s).

$$\begin{bmatrix} T_{xxxx} & T_{xxyy} & T_{xxzz} & T_{xxyz} & 0 & 0 & T_{xxzy} & 0 & 0 \\ T_{xxyy} & T_{xxxx} & T_{xxzz} & -T_{xxyz} & 0 & 0 & -T_{xxzy} & 0 & 0 \\ T_{zzxx} & T_{zzxx} & T_{zzzz} & 0 & 0 & 0 & 0 & 0 & 0 \\ T_{yzxx} & -T_{yzxx} & 0 & T_{yzyz} & 0 & 0 & T_{yzzy} & 0 & 0 \\ 0 & 0 & 0 & 0 & T_{yzzy} & T_{yzxx} & 0 & T_{yzyz} & T_{yzxx} \\ 0 & 0 & 0 & 0 & T_{xxzy} & \frac{T_{xxxx}}{2} - \frac{T_{xxyy}}{2} & 0 & T_{xxyz} & \frac{T_{xxxx}}{2} - \frac{T_{xxyy}}{2} \end{bmatrix}$$

$$T_{xxxx} = \mathbb{T}_0^{(1)} + 2\mathbb{T}_0^{(2)} + 3\mathbb{T}_{40}^{(1)} - 2\mathbb{T}_u^{(1)} - 4\mathbb{T}_u^{(2)}$$
$$T_{xxyy} = \mathbb{T}_0^{(1)} + \mathbb{T}_{40}^{(1)} - 2\mathbb{T}_u^{(1)}$$
$$T_{xxzz} = \mathbb{T}_0^{(1)} - 4\mathbb{T}_{40}^{(1)} + \mathbb{T}_u^{(1)} + 3\mathbb{T}_u^{(3)}$$
$$T_{xxyz} = 2\mathbb{M}_{3c}^{(1)} + \mathbb{M}_{3c}^{(2)} + \mathbb{T}_{4s}^{(1)}$$
$$T_{xxzy} = 2\mathbb{M}_{3c}^{(1)} - \mathbb{M}_{3c}^{(2)} + \mathbb{T}_{4s}^{(1)}$$
$$T_{zzxx} = \mathbb{T}_0^{(1)} - 4\mathbb{T}_{40}^{(1)} + \mathbb{T}_u^{(1)} - 3\mathbb{T}_u^{(3)}$$
$$T_{zzzz} = \mathbb{T}_0^{(1)} + 2\mathbb{T}_0^{(2)} + 8\mathbb{T}_{40}^{(1)} + 4\mathbb{T}_u^{(1)} + 8\mathbb{T}_u^{(2)}$$
$$T_{yzxx} = -2\mathbb{M}_{3c}^{(1)} + \mathbb{T}_{4s}^{(1)}$$
$$T_{yzyz} = \mathbb{T}_0^{(2)} - 4\mathbb{T}_{40}^{(1)} + \mathbb{T}_u^{(2)} - 3\mathbb{T}_u^{(5)}$$
$$T_{yzzy} = \mathbb{T}_0^{(2)} - 4\mathbb{T}_{40}^{(1)} + \mathbb{T}_u^{(2)} + 3\mathbb{T}_u^{(5)}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} T_{yzxx} & -T_{yzxx} & 0 & T_{yzyz} & 0 & 0 & T_{yzzy} & 0 & 0 \\ 0 & 0 & 0 & 0 & -T_{yzzy} & -T_{yzxx} & 0 & -T_{yzyz} & -T_{yzxx} \\ 0 & 0 & 0 & 0 & 0 & T_{xyxy} & 0 & 0 & -T_{xyxy} \end{bmatrix}$$

$$T_{yzxx} = \mathbb{M}_{3c}^{(3)}$$
$$T_{yzyz} = \mathbb{T}_0^{(3)} - 2\mathbb{T}_u^{(4)} - 3\mathbb{T}_u^{(6)}$$
$$T_{yzzy} = -\mathbb{T}_0^{(3)} + 2\mathbb{T}_u^{(4)} - 3\mathbb{T}_u^{(6)}$$
$$T_{xyxy} = \mathbb{T}_0^{(3)} + 4\mathbb{T}_u^{(4)}$$

* Rank 4 tensor (t).

$$\begin{bmatrix} T_{xxxx} & 0 & 0 \\ 0 & T_{xxxx} & T_{yyyz} \\ 0 & 0 & T_{zzzz} \\ 0 & T_{yyyz} & T_{yyzz} \\ T_{zzxx} & 0 & 0 \\ 0 & T_{xxyy} & -T_{yyyz} \\ 0 & T_{yzzy} & 0 \\ 0 & -T_{yyyz} & -\frac{T_{xxxx}}{2} + \frac{3T_{xxyy}}{2} + T_{yyzz} - T_{yzzy} + T_{zzxx} \\ \frac{T_{xxxx}}{2} - \frac{T_{xxyy}}{2} & 0 & 0 \\ -T_{yyyz} & 0 & 0 \end{bmatrix}$$

$$T_{xxxx} = 3\mathbb{T}_0^{(1)} + 3\mathbb{T}_{40}^{(1)} - 2\mathbb{T}_u^{(1)} - 4\mathbb{T}_u^{(2)}$$
$$T_{yyyz} = -\mathbb{T}_{4s}^{(1)}$$
$$T_{zzzz} = 3\mathbb{T}_0^{(1)} + 8\mathbb{T}_{40}^{(1)} + 4\mathbb{T}_u^{(1)} + 8\mathbb{T}_u^{(2)}$$
$$T_{yyzz} = \mathbb{T}_0^{(1)} - 4\mathbb{T}_{40}^{(1)} + \mathbb{T}_u^{(1)} + 3\mathbb{T}_u^{(3)}$$
$$T_{zzxx} = \mathbb{T}_0^{(1)} - 4\mathbb{T}_{40}^{(1)} + \mathbb{T}_u^{(1)} - 3\mathbb{T}_u^{(3)}$$
$$T_{xxyy} = \mathbb{T}_0^{(1)} + \mathbb{T}_{40}^{(1)} - 2\mathbb{T}_u^{(1)}$$
$$T_{yzzy} = \mathbb{T}_0^{(1)} - 4\mathbb{T}_{40}^{(1)} + \mathbb{T}_u^{(2)} + 3\mathbb{T}_u^{(5)}$$